Section 1: Identification

- **Product Name:** Solo AI
- **Models Covered:** Solo AI (Base Model), Solo AI (w/ TOTL PC)
- **Product Type:** Autonomous robotic system
- **Manufacturer / Supplier:** Trossen Robotics
- **Recommended Use:** Research, development, robotics education, and integration into robotic systems
- **Address:** 2854 Hitchcock Ave, Downers Grove, IL, 60515, United States
- **Phone: +**1(630)437-5128
- **Emergency Phone Number:** CHEMTREC (24-hour): +1(800)424-9300
- **Email:** sales@trossenrobotics.com
- **Support:** https://www.trossenrobotics.com/support
- **Revision Date:** 2026-02-04

Section 2: Hazards Identification

2.1 Classification of the Substance or Mixture

This product is an article and is not classified as hazardous according to the Globally Harmonized System (GHS). No hazardous substances are released under normal conditions of use.

2.2 GHS Label Elements

Signal Word: None
Hazard Pictograms: None
Hazard Statements: None
Precautionary Statements: None

2.3 Other Hazards

Although not classified as hazardous under GHS, the product may present **non-chemical hazards** associated with its electromechanical operation, installation, servicing, misuse, or abnormal conditions. These hazards are not subject to GHS classification but are identified for safety awareness and risk management purposes.

Potential non-chemical hazards include:

- **Electrical Shock:** Risk of electric shock from power supplies, wiring, or internal components if the product is improperly installed, damaged, or serviced while energized.
- **Mechanical Hazards / Moving Parts:** Risk of injury from moving components, including pinch points, rotating elements, or articulated mechanisms during operation or maintenance.

- **Unexpected Motion (Autonomy):** Autonomous or semi-autonomous operation may result in unanticipated movement if safety interlocks, sensors, or software controls are bypassed, misconfigured, or fail.
- **Optical / Sensor Exposure:** Optical / Sensor Exposure: The Solo AI system incorporates camera and optical sensing devices, including a camera integrated with the WidowX AI Follower and an additional external camera mounted on a fixed stand. Direct exposure does not present a chemical hazard; however, removal of protective housings or misuse may pose minor optical exposure risks.
- **Hot Surfaces:** Motors, actuators, or electronic components may become warm during extended operation.

These hazards are mitigated through product design, protective enclosures, software controls, labeling, and operating instructions when the product is used as intended.

2.4 Hazards Not Otherwise Classified (HNOC)

None known.

Section 3: Composition / Information on Ingredients

3.1 Substances

Not applicable. This product is an article.

3.2 Mixtures

Not applicable.

3.3 Articles

This product is an assembled electromechanical article and does not release hazardous chemicals under normal conditions of use.

The WidowX AI product family consists of robotic arms, actuators, electronic controllers, wiring, sensors, and accessories that are fully enclosed or integrated during normal operation.

Components may include, but are not limited to:

- Electric motors and actuators
- Electronic circuit boards and controllers
- Power distribution and control electronics
- Sensors and encoders
- Camera and vision equipment (Follower model only)
- Camera mounting hardware and fixed camera stands
- Structural metal and polymer components
- Cables, connectors, and fasteners

These components do not present chemical hazards under normal conditions of use and are not classified as hazardous under GHS.

Additional Information

No components are present above reportable thresholds requiring disclosure under GHS, OSHA Hazard Communication, or equivalent international regulations.

Section 4: First Aid Measures

4.1 Description of First-Aid Measures

General Advice:

No chemical exposure, inhalation, eye contact, or ingestion is expected under normal conditions of use. If an accident occurs, provide first aid appropriate to the nature of the injury and seek medical attention as indicated.

Inhalation:

Not applicable under normal use. If smoke or fumes are inhaled due to electrical failure or fire, move to fresh air and seek medical attention if symptoms persist.

Skin Contact:

Not applicable under normal use. Treat mechanical injuries or thermal burns per standard first-aid procedures.

Eye Contact:

Flush eyes with clean water if exposed to debris or particulate matter. Seek medical attention if irritation persists.

Ingestion:

Not applicable.

4.2 Most Important Symptoms and Effects

- Electrical shock
- Mechanical injury
- Thermal burns

4.3 Indication of Immediate Medical Attention

Immediate medical attention is required in cases of:

- Electrical shock
- Significant mechanical injury
- Burns

Section 5: Fire-Fighting Measures

5.1 Suitable Extinguishing Media

- Carbon dioxide (CO_2)
- Dry chemical powder
- Fire-fighting foam
- Water spray or fog (for cooling and fire control)

5.2 Specific Hazards Arising from the Product

- Combustion of plastic, polymer, and electronic components may release hazardous fumes
- Energized electrical components may present additional hazards during fire conditions

5.3 Advice for Firefighters

- Wear self-contained breathing apparatus (SCBA) and full protective fire-fighting gear
- If safe to do so, disconnect electrical power sources before firefighting activities

Section 6: Accidental Release Measures

6.1 Personal Precautions

No chemical release is expected under normal use.

In the event of mechanical damage, electrical failure, or fire:

- De-energize equipment if safe to do so
- Avoid contact with damaged electrical components
- Use appropriate PPE during cleanup

6.2 Environmental Precautions

Prevent debris or damaged components from entering drains or waterways.

6.3 Methods for Cleanup

Dispose of damaged components in accordance with local regulations and e-waste requirements.

Section 7: Handling and Storage

7.1 Handling

- Operate the product only in accordance with manufacturer instructions
- Disconnect power before servicing or maintenance
- Do not open sealed electrical enclosures.
- Avoid crushing, puncturing, or subjecting the product to mechanical shock
- Use proper lifting techniques when transporting the system

7.2 Storage

- Store in a dry, well-ventilated area
- Avoid exposure to moisture, corrosive environments, or direct sunlight
- Store within recommended temperature ranges specified by the manufacturer

Section 8: Exposure Controls / Personal Protection

8.1 Control Parameters

- Occupational exposure limits: Not established / Not applicable
- No chemical exposure is expected during normal operation

8.2 Exposure Controls

Engineering Controls:

- Protective enclosures
- Software interlocks and safety controls
- Electrical insulation and grounding

Personal Protective Equipment (PPE):

- **Eye Protection:** Safety glasses during servicing or maintenance
- **Hand Protection:** Protective gloves when handling components
- **Skin Protection:** Not required under normal use
- **Respiratory Protection:** Not required under normal use. Required if fumes are present due to electrical fire or component overheating.

Hygiene Measures

- Wash hands after servicing
- Do not eat, drink, or smoke while performing maintenance

Section 9: Physical and Chemical Properties

9.1 Information on Basic Physical and Chemical Properties

- **Physical State:** Solid article
- **Appearance:** As manufactured (assembled robotic system)
- **Odor:** None
- **pH:** Not applicable
- **Melting / Freezing Point:** Not applicable
- **Boiling Point:** Not applicable
- **Flash Point:** Not applicable
- **Evaporation Rate:** Not applicable
- **Flammability (Solid/Gas):** Not flammable as supplied
- **Upper/Lower Flammability Limits:** Not applicable
- **Vapor Pressure:** Not applicable
- **Vapor Density:** Not applicable
- **Relative Density:** Not applicable
- **Solubility:** Insoluble in water
- **Auto-ignition Temperature:** Not applicable
- **Decomposition Temperature:** Not determined

9.2 Other Information

Components may generate heat during operation.

Section 10: Stability and Reactivity

10.1 Reactivity

No hazardous chemical reactivity is expected under normal conditions of use.

10.2 Chemical Stability

Stable under normal operating, handling, and storage conditions.

10.3 Possibility of Hazardous Reactions

No hazardous reactions expected during normal use.

10.4 Conditions to Avoid

- Water ingress into electrical systems
- Mechanical damage, crushing, or puncturing
- Exposure to temperatures outside manufacturer specifications

10.5 Incompatible Materials

None known under normal use.

10.6 Hazardous Decomposition Products

Combustion byproducts of plastics and electronic components in the event of fire.

Section 11: Toxicological Information

11.1 Information on Likely Routes of Exposure

- Inhalation: Not applicable under normal use
- Skin contact: Not applicable under normal use
- Eye contact: Not applicable under normal use
- Ingestion: Not applicable

11.2 Symptoms Related to Physical, Chemical, and Toxicological Characteristics

No toxicological effects are expected during normal operation. Potential adverse effects may occur only under abnormal conditions, including smoke or fume inhalation during overheating or fire.

11.3 Information on Toxicological Effects

This product is an article and does not release hazardous substances under normal conditions.

Section 12: Ecological Information

12.1 Toxicity

No known environmental toxicity under normal conditions of use.

12.2 Persistence and Degradability

Not applicable for an assembled article.

12.3 Bioaccumulative Potential

Not applicable.

12.4 Mobility in Soil

Not applicable.

12.5 Results of PBT and vPvB Assessment

Not applicable.

12.6 Other Adverse Effects

Improper disposal of electronic components may cause environmental harm.

Section 13: Disposal Considerations

13.1 Waste Treatment Methods

- Dispose of electronic components in accordance with local, regional, and national e-waste regulations.
- Do not incinerate electronic components.

Follow all applicable regulations, including WEEE, EPA, and equivalent international directives.

Section 14: Transport Information

This product is not regulated as a dangerous good for transport by road, rail, sea, or air when shipped as an assembled system.

Section 15: Regulatory Information

15.1 Safety, Health, and Environmental Regulations

- **OSHA Hazard Communication Standard:** Not classified as hazardous
- **GHS:** Not classified
- **REACH:** Article; no SVHC above reporting thresholds
- **RoHS:** Compliant (where applicable)
- **California Proposition 65:** This product may contain trace substances subject to Prop 65 requirements.

This product complies with applicable electrical, EMC, and safety regulations where sold.

Section 16: Other Information

Referenced SDS

Manufacturer Safety Data Sheets are available upon request.

SDS Metadata (Recommended for Audit Readiness)

- **SDS Preparation Date:** 2026-02-04
- **Revision Date:** 2026-02-04
- **Revision History:** Initial release

Disclaimer

This Safety Data Sheet is provided for informational purposes to support safe handling and use of the product. It does not replace product manuals, operating instructions, or applicable regulatory requirements.

Contact for SDS Information

Trossen Robotics
Email: sales@trossenrobotics.com
Support: https://www.trossenrobotics.com/support